



COSC 2436 | Programming and Data Structures

Section: 12348 | Face-to-Face

Prerequisite(s): A grade of C- or better in COSC 1437 and MATH 2305, and concurrent enrollment in or credit for MATH 2414 with a grade of C- or better. Introduction to fundamental data structures: arrays, lists, stacks, queues, hash tables, trees; sorting and searching; graph algorithms; design, analysis, and comparison of algorithms. Correctness verification techniques such as assertions and invariants. Review program specification, unit testing, and debugging.

Instructor Information

Dr. Nouhad Rizk

Department: Computer Science Department

Email: njrizk@uh.edu

Phone: 713/743-3710

Office Hours

12:00 PM - 1:00 PM Tuesdays and Thursdays

General Course Information

(Optional: Enter Secondary Course Title)

Course Meeting Time: (Enter your class time)

Course Meeting Location: (Enter location or Delete this line)

Course Objectives and Student Learning Outcomes

Program Objectives

The objective is to prepare students such that within a few years of graduation they will be able to secure for themselves a successful computer science career and contribute decisively to the improvement and development of technology by demonstrating their ability to:

1. Address and solve complex broadly defined problems related to their discipline and field of specialization.
2. Work as team members, show leadership, and communicate technical concepts and ideas effectively.
3. Manifest a high level of professional integrity and make ethical decisions that will have a positive impact on the organization and society.
4. Embrace and practice lifelong learning, continue personal growth, and professional self- improvement.

Student Learning Outcomes

Based on satisfactory completion of the course, a student should be able to:

- Demonstrate an understanding of basic data structures (such as an array-based list, linked list, stack, queue, binary search tree) and algorithms.
- Demonstrate the ability to analyze, design, apply and use data structures and algorithms to solve problems and evaluate their solutions.
- Demonstrate an understanding of analysis of algorithms. Study an algorithm or program code segment that contains iterative constructs and analyze the asymptotic time complexity of the algorithm or code segment.

Required Instructional Materials

Text Book:

Required: Data Structures and Algorithms in C++, by M. T. Goodrich, R. Tamassia, and D. Mount, 2nd ed., 2011, Wiley, ISBN 13-978-0-470-38327-8. (Any C++ reference book may be helpful as well.)

Open Book: <https://open.umn.edu/opentextbooks/textbooks/open-data-structures-an-introduction> Reference:

- Malik, D S. Data Structures in C++, 2nd Edition, Cengage Learning

- Weiss, Mark A. Data Structures and Algorithm Analysis in C++. 4th Edition. Pearson 2014. ISBN-13: 978-0-13-284737-7

Compiler:

Microsoft Windows Visual Studio 15 or higher GCC 5.3 for Linux

Clang or g++ compilers for OS X (check online for latest versions)

Course Schedule, Assignments, and Assessments

Date	Topic	Reading	Extras
6/8	Basic C++ Recursion	Chapter 1 Chapter 2	C 1.7 R 2.12 C 2.4
6/9	Array-Based Implementations Link-Based Implementations	Chapter 3	C 3.7 C 3.8 S 3.14
6/10	Introduction of Algorithms	Chapter 4	C 4.5 C 4.23
6/11	Analysis of Algorithms LAST DAY TO DROP WITHOUT W	Chapter 4	C 4.26
6/12	Doubly linked List		

6/13	NO CLASS (Weekend)		
6/14	NO CLASS (Weekend)		
6/15	EXAM 1 - WORTH 17%		

6/16	Stacks Queues	Chapter 5	C 5.2 C 5.5 C 5.6
6/17	Bubble sort Selection sort	Chapter 6	C 6.12 C 6.17
6/18	Insertion sort Priority Queues Heaps	Chapter 6, 8	C 6.20, C8.4, C8.5, C8.17
6/19	OFF		
6/20	NO CLASS (Weekend)		
6/21	NO CLASS (Weekend)		
6/22	EXAM 2 - WORTH 17%		
6/23	Hashing	Chapter 9	C 9.12 C 9.13 C 9.14
6/24	Trees Binary Tree Traversal	Chapter 7	C 7.4 C 7.5 C 7.34
6/25	BST	Chapter 10	C 10.2 C 10.9
6/26	AVL Trees	Chapter 10	C 10.10
6/27	NO CLASS (Weekend)		
6/28	NO CLASS (Weekend)		
6/29	EXAM 3 - WORTH 17%		
	Last day to drop with W		

6/30	Divide and Conquer Sorting	Chapter 11	C 11.12 C 11.13 C 11.18
7/1	Graph Graph Traversals	Chapter 13	C 3.5 C 3.8
7/2	Minimum Spanning Tree – Prim, Kruskal	Chapter 13	C 3.26
7/3	OFF		C 3.26
7/4	NO CLASS (Weekend)		
7/5	NO CLASS (Weekend)		
7/6	Shortest Path – Dijkstra	Chapter 13	C 3.26
7/7	Tuesday EXAM 4 - WORTH 17%		
7/8	NO CLASS Deadline for hw regrade		

Discussion and Lecture Topics

ALL HWs, DLabs, and extras should be submitted to the server.

Your username is posted under the Grades section in Canvas.

1. your **username** is as your cougarnet name, the hostname of the server as 2436.cs.uh.edu.
2. The default password for each account is the word password with the last 4 digits of people soft id (no space). (Password would be **passwordxxxx**.)
3. Please, change your password as soon as you logged in.

Homework: Homework must be submitted to the server. Each homework has two regrades.

	Time	Max Credit
Initial Grade:	Due Date	100%
1st Regrade:	24 h	80%
2nd Regrade:	48 h	60%

- The first regrade is the original score + (the score after 1st regrade - original score) * 80%. For example, if you original grade was a 40 and then you got an 80 after the first regrade, then your grade after the first regrade would be $40 + (80 - 40) * 80\% = 72$.
- The second regrade is the score after the first regrade + (score after the second regrade - score after the first regrade) * 60%. For example, if your grade after the first regrade was a 72 (as in the example above) and then you got a 100 on the second regrade, your final homework grade would be $72 + (100 - 72) * 60\% = 88$.
- Your final homework grade can be calculated by doing: $(\text{original_grade} + (\text{1st_regrade_grade} - \text{original_grade}) * 80\%) + (\text{2nd_regrade_grade} - \text{grade_after_1st_regrade}) * 60\%$

To submit a regrade, update the homework directory with the new program files before the time listed above. Results of your attempt are stored in a “-graded” directory.

Final Regrade

In addition, students are given the chance to regrade their lowest scoring homework assignment for 100% credit at the end of the semester. Be sure to update your files before July 1st.

Physical Lab TuTh 12:00PM - 2:00PM Room S205 and Dlab Rules



<https://www.cses.fi/problemset/>

Attending the lab every Tuesday and Thursday is mandatory. You are expected to work on your DLab selection during these sessions.

1. As soon as you enter the room, proceed directly to your assigned group. Groups are organized alphabetically by the first letter of your last name. For example, Eric Davila would join Group D. Some groups may cover multiple letters (e.g., Group I_J for last names starting with I or J). Groups will be created in the Lab section on Canvas by ORD day.
2. Group leaders will rotate alphabetically and are responsible for recording group attendance in the provided Excel sheet each day. The sheet (e.g., "Attendance_Tuesday_June_3") must include each member's PeopleSoft ID and attendance status (1 = present, 0 = absent). The completed sheet should be submitted to the Lab TA via email or in person the same day.
3. In each session, the group will collaboratively work on a selected DLab problem from one of its members. While DLabs may be completed collaboratively, each student must submit their own work individually (**Gantt Chart**). Before leaving the lab, all students are required to update their personal Gantt chart based on their progress.

Group	Last name starting with
1	A,B
2	C
3	D,E
4	F
5	G,H,I,J
6	K,L,M
7	N,O,P,Q,R
8	S
9	T
10	U,V,W,X,Y,Z

Grading Rubrics and Weights

The final numeric grade is computed based on the student's performance in weekly assignments and exams. The final numeric grade for the course will be determined as follows:

- **Exams (4 total) 68%**
 - Exam1 Monday June 15th,in class at the same time as the lecture 10:00 AM -12:00 PM.
 - Exam2 Monday June 22th,in class at the same time as the lecture 10:00 AM -12:00 PM.
 - Exam3 Monday June 29th,in class at the same time as the lecture 10:00 AM -12:00 PM.
 - Exam4 Tuesday July 7,in class at the same time as the lecture 10:00 AM -12:00 PM.
- **Homework assignments (4 total) 12%**
 - hw0 due Tuesday June 8 ,not graded but submission is required.
 - hw1 due Sunday June 14 at 11:59 PM.
 - hw2 due Sunday June 21th at 11:59 PM.
 - hw3 due Sunday June 28th at 11:59 PM.
 - hw4 due Sunday July 5th at 11:59 PM.
- **DLabs: Each Dlab is 3 selected problems Dlab1_1, Dlab1_2,Dlab1_3 (3 total) 10%**
 - Dlab1 due Sunday June 21st at 11:59 PM
 - Dlab2 due Sunday June 28st at 11:59 PM
 - Dlab3 due Sunday July 5th at 11:59 PM
- **Glabs (3 Total) 0% [Optional for each Dlab: Each Glab is 2 Generated problems]**
 - Glab1 (two problems like Dlab1) due Sunday June 21st at 11:59 PM.
 - Glab2 (two problems like Dlab2) due Sunday June 28th at 11:59 PM.
 - Glab3 (two problems like Dlab3) due Sunday July 5th at 11:59 PM.
- **CMAPS (4 total) 4%**
Each submission should be in. Cxl and .pdf. A CMAP algorithm can be of a hw or an extra
 - CMAP1 due Saturday June 13th at 11:59 PM

- CMAP2 due Saturday June 20th at 11:59 PM
- CMAP3 due Saturday June 27th at 11:59 PM
- CMAP4 due Sunday July 5th at 11:59 PM
- **CIA (ChatGPT Interactive Activities) - Group of 3 (2 total) 3%**
The purpose is to develop 20 questions with or without ChatGPT. Rules: A grade of zero shall be given if a violation of any of the following rules is found:
 - The number of questions should be balanced between the different types of questions.
 - Fill in the blank should cover all possibilities
 - Straight forward questions will not be accepted in any type especially true/false.
 - Essai's questions should be specific with the submission of a detailed answer.
(Explain topic 1is not acceptable)
- CIA1 due Tuesday June 16th at 11:59 PM
- CIA2 due Tuesday June 23rd at 11:59 PM
- **Attendance Lecture and Lab 3%**
- **Gantt Chart in the lab will be additional 1.5% (100+1.5)**
- **Extra problems 0%**

Extra problems may not directly improve your grade. These serve to boost averages on the lower end of a letter grade. Extra problems are optional; you can do up to 5 per week from the suggested problems in the syllabus and from the end of chapters. You can also upload additional programs that exercise the material learned in class as extra. To upload an extra, create a new directory under your root directory with the month, week, and number. Each week, the numbering starts at 1. For example, label the two extras of week 1, "extra_6_1_1" and "extra_6_1_2". Next week, you can add "extra_6_2_1".

A >= 92.5	A- >= 89.5 and < 92.5	B+ >= 86.5 and < 89.5
Excellent	Outstanding	Very Good
B >= 83.5 and < 86.5	B- >= 79.5 and < 83.5	C+ >= 76.5 and < 79.5
Good	Above Average	High Average

C ≥ 72.5 and < 76.5 Average	C- ≥ 69.5 and < 72.5 Low Average	D+ ≥ 65.5 and < 69.5 Below Average
D ≥ 62.5 and < 65.5 Poor	F < 62.5 Failing	

Course Policies and Procedures

No-late submission policy

No-exam make up policy

Important Notes

1. 1-Day Policy: One has 1 day starting from the end of the class time in which the graded assignment/exam papers have been distributed and/or posted in order to object to the score of that assignment or exam. The objection shall be submitted electronically by emailing the TA and the instructor.
2. Academic Honor Code: As a student, you join a community of scholars who are committed to excellence in learning. I assume that students will pursue their studies with integrity and honesty. zero- tolerance for cheating, whether in exams, quizzes or programming assignments. Plagiarism, copying and other anti- intellectual behavior are prohibited by the university regulations. Violators will face serious consequences.
3. Student Conduct: Disruptive behavior inside or outside class may result in disciplinary actions and academic failure. Students must refrain from disturbing the peace and good order of the university. For more details, please refer to <http://www.uh.edu/dos/pdf/codeofconduct.pdf>
4. Academic Integrity: Cheating or any other suspected violations of academic integrity will not be tolerated and will be reported to the Department of Computer Science, Director of Undergraduate/Graduate Studies and if substantiated may result in significant penalty. It is each student's responsibility to read and understand the Academic Honesty Policy found in the Student Handbook (<http://www.uh.edu/academics/catalog/policies/academ-reg/academic-honesty/>).
5. Plagiarism: Plagiarism is using someone else's work without proper acknowledgement. This includes getting help from a friend or colleague and online material. When using someone else's work, always cite the source. Plagiarism is

considered a serious breach of academic integrity. Any breach of academic integrity or plagiarism would result in a minimum of one full letter grade reduction over the final score and possible expulsion from university.

6. Attendance: Attendance is strongly recommended. It will not be checked, but it might be considered in borderline decisions for the final grade. Students with unexcused absences should not expect additional help outside of class and are still responsible for any material or instructions given in class, for turning in assignments on time, and for taking exams at the scheduled times.
7. Make Up: NO Make up policy. Missed exams might only be considered (the excused absence does not guarantee exam scheduling) for excused absences with advanced notice (BEFORE THE EXAM) of at least two working days with full documentation. A grade of zero will be assigned for missed exams, if documentation has been submitted within one day or the day of the exam or after the exam date.
8. Reading: Readings from the textbook will be given out throughout the semester. We will not be able to discuss all of it in class, but you are still responsible for keeping up with these readings.
9. Code Documentation: When assignments are graded, source code may be examined to verify the way a solution was achieved or to award partial credit. It is your responsibility to make sure that your source code is presented in a clear, readable, way.
10. NO LATE POLICY ON ANYTHING=> nothing will be accepted late!

Classroom rules:

- Listen actively and attentively.
- Do not use headphones or earbuds, this is very offending. The instructor can ask you to leave the classroom.
- Ask for clarification if you are confused.
- Do not interrupt one another.
- Challenge one another but do so respectfully.
- Critique ideas, not people.
- Avoid put-downs (even humorous ones).
- Take responsibility for the quality of the discussion.
- Build on one another's comments; work toward shared understanding.
- Do not monopolize discussion.
- Speak from your own experience, without generalizing.

- If you are offended by anything said during discussion, acknowledge it immediately.
- Consider anything that is said in class strictly confidential.

University Policies and Student Support Resources

Mental Health and Wellness Resources

The University of Houston has a number of resources to support students' mental health and overall wellness, including [CoogsCARE](#) and the [UH Go App](#). [UH Counseling and Psychological Services \(CAPS\)](#) offers 24/7 mental health support for all students, addressing various concerns like stress, college adjustment and sadness. CAPS provides individual and couples counseling, group therapy, workshops and connections to other support services on and off- campus. For assistance visit uh.edu/caps, call 713-743-5454, or visit a [Let's Talk](#) location in-person or virtually. Let's Talk are daily, informal confidential consultations with CAPS therapists where no appointment or paperwork is needed.

Need Support Now? If you or someone you know is struggling or in crisis, help is available. Call CAPS crisis support 24/7 at 713-743-5454, or the National Suicide and Crisis Lifeline: call or text 988, or chat 988lifeline.org.

Title IX/Sexual Misconduct

Per the UHS Sexual Misconduct Policy, your instructor is a "responsible employee" for reporting purposes under Title IX regulations and state law and must report incidents of sexual misconduct (sexual harassment, non-consensual sexual contact, sexual assault, sexual exploitation, sexual intimidation, intimate partner violence, or stalking) about which they become aware to the Title IX office (known at UH as the Equal Opportunity Services office or "EOS"). Please know there are places on campus where you can make a report in confidence. You can find more information about resources on the UH [Title IX/Sexual Misconduct Resources page](#). Please note that you may also report concerns of discrimination based on your protected class identity to EOS.

Reasonable Academic Adjustments/Auxiliary Aids

The University of Houston is committed to providing an academic environment and educational programs that are accessible for its students. Any student with a disability who is experiencing barriers to learning, assessment or participation is encouraged to contact the Justin Dart, Jr. Student Accessibility Center (Dart Center) to learn more about academic accommodations and support that may be available to them. Students seeking academic accommodations will need to register with the Dart Center as soon as possible to ensure timely implementation of approved accommodations. Please contact the Dart Center by visiting the website: <https://uh.edu/accessibility/> calling (713) 743-5400, or emailing jdcenter@Central.UH.EDU.

The [Student Health Center](#) offers a Psychiatry Clinic for enrolled UH students. Call 713-743-5149 during clinic hours, Monday through Friday 8 a.m. - 4:30 p.m. to schedule an appointment.

The [A.D. Bruce Religion Center](#) offers spiritual support and a variety of programs centered on well-being.

The [Center for Student Advocacy and Community \(CSAC\)](#) is where you can go if you need help but don't know where to start. CSAC is a "home away from home" and serves as a [resource hub](#) to help you get the resources needed to support academic and personal success. Through our [Cougar Cupboard](#), all students can get up to 30 lbs of FREE groceries a week. Additionally, we provide 1:1 appointments to get you connected to on- and off-campus resources related to essential needs, safety and advocacy, and more. The [Cougar Closet](#) is a registered student organization advised by our office and offers free clothes to students so that all Coogs can feel good in their fit. We also host a series of cultural and community-based events that fosters social connection and helps the cougar community come closer together. Visit the CSAC homepage or follow us on Instagram: @uh_CSAC and @uhcupbrd. YOU belong here.

Cougar Parent and Family Engagement Office

This office, under the Dean of Students Office (<https://www.uh.edu/dos/>), coordinates resources for Pregnant and Parenting Students (<https://www.uh.edu/dos/parenting-students/>), Parents and Family members of UH students (<https://www.uh.edu/dos/parents/>), and Sexual Misconduct Support Services

(<https://www.uh.edu/dos/sexual-misconduct-support-services/>). A description of available support and resources are provided on the respective web pages. The Cougar Parent and Family Engagement Office is located in Student Center South, room B12.

Academic Honesty Policy

High ethical standards are critical to the integrity of any institution, and bear directly on the ultimate value of conferred degrees. All UH community members are expected to contribute to an atmosphere of the highest possible ethical standards. Maintaining such an atmosphere requires that any instances of academic dishonesty be recognized and addressed. The [UH Academic Honesty Policy](#) is designed to handle those instances with fairness to all parties involved: the students, the instructors, and the University itself. All students and faculty of the University of Houston are responsible for being familiar with this policy.

Excused Absence Policy

Regular class attendance, participation, and engagement in coursework are important contributors to student success. Absences may be excused as provided in the University of Houston [Undergraduate Excused Absence Policy](#) and [Graduate Excused Absence Policy](#) for reasons including medical illness of student or close relative, death of a close family member, legal or government proceeding that a student is obligated to attend, recognized professional and educational activities where the student is presenting, and University-sponsored activity or athletic competition. Under these policies, students with excused absences will be provided with an opportunity to make up any quiz, exam or other work that contributes to the course grade or a satisfactory alternative. Please read the full policy for details regarding reasons for excused absences, the approval process, and extended absences. Additional policies address absences related to [military service](#), [religious holy days](#), [pregnancy and related conditions](#), and [disability](#).

Recording of Class

Students may not record all or part of class, livestream all or part of class, or make/distribute screen captures, without advanced written consent of the instructor. If you have or think you may have a disability such that you need to record class-related activities, please contact the [Justin Dart, Jr. Student Accessibility Center](#). If you have an accommodation to record class-related activities, those recordings may not be shared with

any other student, whether in this course or not, or with any other person or on any other platform. Classes may be recorded by the instructor. Students may use instructor's recordings for their own studying and notetaking. Instructor's recordings are not authorized to be shared with anyone without the prior written approval of the instructor. Failure to comply with requirements regarding recordings will result in a disciplinary referral to the Dean of Students Office and may result in disciplinary action.